Assignment 6 - Advanced Dashboard UX & Parameter Passing

Target Audience: B.Tech Computer Science **Total Marks:** 10

Objective

Architect a multi-tab executive dashboard utilizing complex layout containers, in-dashboard custom calculations, and multi-parameter cross-dashboard drill-throughs.

Dataset

Source: Global Video Game Sales **URL:** Video Game Sales (CSV) *(Ingest and create a module)*.

Task Requirements (No-Step Instructions)

1. **Dashboard Layout & UI (Tab 1):**
   - Create a Dashboard. Use a layout that includes a fixed side-navigation bar or top banner.
   - Tab 1 ("Executive Overview"): Use the AI Assistant to generate a visualization showing "Global Sales by Genre".
   - Create a custom calculation *directly within the dashboard interface* (not the data module). Calculation: `[Global_Sales] - [NA_Sales]` . Name it "Rest of World Sales" and display it as a KPI widget.
2. **Conditional Alerting (Tab 1):**
   - Apply conditional formatting to the "Rest of World Sales" KPI. If the value > 500, highlight the background Green. If < 500, highlight Red.
3. **Detail Construction & Global Filters (Tab 2):**
   - Tab 2 ("Publisher Details"): Build a Scatter Plot mapping `NA_Sales` vs `EU_Sales` , colored by `Publisher` . Apply a Top 20 filter to `Publisher` by Global Sales.
   - Add a Global Filter widget (All Tabs) for `Platform` .
4. **Multi-Parameter Drill-Through:**
   - Create a Bar Chart on Tab 1 showing Sales by `Year` .
   - Configure a Drill-Through definition from the Genre chart on Tab 1 to Tab 2.
   - *Complexity requirement:* The drill-through must be configured so if a user selects a Genre, it filters Tab 2, AND if they also have a `Platform` selected in the global filter, both contexts pass to the target.
5. **Finalization:** Save as `Gaming_Advanced_Dashboard` .

Grading Rubric (10 Marks)

- **2 Marks:** Implementation of advanced layout, AI utilization, and UI aesthetics.
- **3 Marks:** Successful creation of the in-dashboard custom calculation (Rest of World Sales).
- **2 Marks:** Accurate conditional formatting rules applied to the KPI.
- **3 Marks:** Flawless configuration of the complex drill-through passing multiple contexts (Genre/Platform).